IKP: PQC Migration (2)

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Agreements

The following sentences stand on their own. These are candidate agreements and reported points of convergence for working-group review. Inclusion does not establish consensus, verify a technical claim, or record formal adoption. The working group must confirm each decision before it is cited as such. Claims that were only discussed, including figures and alternatives that were not adopted, remain in full under Discussion.

Encryption migration is driven by harvest-now-decrypt-later and by data lifetime. The federal dates on the table are 2030 for key establishment and 2031 for signatures, on high-impact and high-value systems, with deprecated and disallowed as the two status words.

Four signature standards are the current set: SLH-DSA (FIPS 205), ML-DSA (FIPS 204), LMS and XMSS (SP 800-208). FN-DSA is selected and not finished. Hybrids are allowed and not mandated. SP 800-227 is the KEM and hybrid key-establishment guidance, including module validation.

ML-DSA is the general-purpose lattice signature. SPHINCS+ is not a general-purpose performance replacement. Extra on-ramp schemes are a contingency for assumption diversity, or a fit for unusual size constraints.

There is no fixed "cut N per round" rule. The on-ramp started at 69. Assumption diversity is intentional. UOV was kept as a class.

Use-case-specific SPHINCS variants are happening and are not NIST standards. Feedback belongs on the PQC Forum, at the NIST PQC conference, in ordinary papers, and sometimes at IETF.

NCCoE is the migration-guidance program (inventory, then plan, sector by sector). The competition repository should point at it. A blockchain use-case category is a useful contribution if the taxonomy has a slot.

Hybrid composition needs metrics and a re-verification loop after optimization. Stacking every algorithm is not a design. Dual post-quantum hybrids are in the question set.

Q-day will force a late merge. Adopting a lattice scheme that fails in that same window is worse than shipping conservative SLH-DSA first and keeping a smaller lattice scheme as the everyday tool afterward.

Hash-based designs are the conservative default under discussion, and they remain an assumption, including for Merkle-tree agility.

Bitcoin and Ethereum differ in governance speed and in signature jobs (consensus votes versus user payments versus Bitcoin's non-reuse norm). Criteria for the competition have to encode that, and they are wanted before the end of the year. Social consensus is harder than the signature math.

Non-migrating and lost coins (10–13% of bitcoin, in the figure used) are part of migration, not a footnote. The disable-versus-wallet-policy choice goes into the report and the evaluation criteria.

sig.golf-style search can shrink size and verification cost, and it can fragment the algorithm set. The competition should be able to say when optimization stops. Key generation on a target device is a constraint, not an afterthought. Signature grinding is acceptable when done correctly.

The git repository is the archive for committee output and SSR papers. The shared note on crypto agility for cryptocurrency, including its Section 4 policy question, is part of that record.

Salvage is a future legal topic for BGIN, not a rule adopted here.

Documents considered

The session referred to the following documents, repositories, and standards. Bibliographic identifications and factual checks added during review are marked as editorial clarifications; they do not imply that the links were spoken or that the meeting adopted the findings. Names of the people who introduced the material are omitted.

Yukon is the public challenge platform linked by ECDSA.fail. The exact signature variant or leaderboard submission discussed still requires its own reference.

`https://github.com/bgin-global/pqc-agility-competition` — competition repository, template stage.

`https://ssresearch26.umbc.edu/` — SSR conference site for the December Baltimore workshop and its PQC workshop call for papers. The captions say SSR, Baltimore, security research, and an extended Monday deadline; they do not speak the URL.

A shared note on crypto agility for cryptocurrency, shared in the chat, with a Section 4 on changing the accepted signing algorithm by policy. No URL was spoken. Do not invent one.

FIPS 205 (SLH-DSA), FIPS 204 (ML-DSA), SP 800-208 (LMS and XMSS), SP 800-227 (KEM recommendations, hybrid key establishment, module validation).

NIST IR 8547, Transition to Post-Quantum Cryptography Standards, initial public draft (2024): `https://csrc.nist.gov/pubs/ir/8547/ipd` Final approval was described as pending in the session.

Deprecated versus disallowed. 2030 and 2031 timelines. Executive Order 14412 was also referenced; the identification does not independently validate every deadline reported in the discussion. An agency memo, untitled in the audio.

PQC Forum mailing list and a pqc-comments address. NIST PQC standardization conference (roughly annual). Crypto (this year's fixed-point Falcon paper).

HQC, in approvals. FN-DSA fixed-point proposal on the PQC Forum.

HAWK: withdrawn on 29 July 2026 after improved cryptanalysis. It was one of the nine selected round-three candidates; its withdrawal reduces that set to eight.

UOV (four schemes kept). MPC-in-the-head, name garbled in the audio.

BIP 360 (Merkle-root, user/wallet choice). Pay-to-Taproot v2 (opt-in future disablement). Proposal 361 (unmigrated coins; poorly received).

sig.golf, the Ethereum auto-research challenge. Lean Ethereum, the formalization roadmap. A sibling challenge aimed at attacking hash-based schemes, name garbled.

ML-KEM, including as one leg of a possible dual post-quantum hybrid with HQC. X-Wing was in the keynote slice, not restated here.

IETF and CFRG, as the venues already drafting hybrid and composite signature behavior. The specific drafts `draft-ietf-lamps-one-signature-certs` and `draft-gray-lamps-composite-fndsa-lms` were not named in the captions. Composite signatures, FN-DSA, and LMS were each discussed, not as that draft. Those filenames are omitted.

NCCoE migration guidance and sector white papers (inventory, then plan). No document titles were spoken.

Session web page, as the place slides would be posted. No URL spoken.

No posters.

Discussion

The following is the substance of the session. It preserves claims, figures, alternatives, and objections. Discussion does not establish a decision. Any candidate agreement requires explicit working-group confirmation.

Repository. At the start of the continuation, the chat was pointed at the git repository BGIN is hosting for the competition. It is still a template. It is the intended aggregation point for research, evaluation-committee outcomes, and papers after the SSR conference.

`https://github.com/bgin-global/pqc-agility-competition`

Federal transition timelines and the current NIST signature set. Harvest-now-decrypt-later was restated as the encryption threat: migration has to finish inside the secrecy lifetime of the data. Federal goals, driven by executive orders and national-security direction, are being written into a transition document. An initial public draft was released in 2024. The final version was hoped for this month or the next (October was the month in view) and was described as stuck in approvals. The

document is to say which algorithms are vulnerable, what the post-quantum replacements are, and what the timelines are. A later executive instrument, number garbled, accelerated the timeline for high-impact systems and high-value assets. Those categories were described as vaguely defined and, in practice, as systems whose compromise would be catastrophic. Federal engineers tend to treat every vulnerability as catastrophic; the federal distinction is breadth of impact. The dates stated: high-value systems migrate key establishment by 2030 and digital signatures by 2031. Other work is due before those dates. An agency memo carries the operational detail.

Two status words: deprecated means still usable, with a known security risk; disallowed means the scheme will no longer be FIPS-validated.

Hybrids are wanted for two different reasons. They are easy: the classical implementation already exists, and signatures are cheap, so the classical signature stays and a post-quantum signature is added. They are also a hedge against an implementation or parameter bug in the post-quantum algorithm, short of a break of the math problem, using a classical scheme customers already trust. Whether a hybrid is used depends on the application and the community. No requirement that hybrids be used was stated. Hybrids are allowed. For key establishment, guidance in SP 800-227 (recommendations for key-encapsulation mechanisms) is what lets a hybrid KEM receive module validation.

Signature algorithms described as standardized now, four of them:

SLH-DSA, FIPS 205.

ML-DSA, FIPS 204.

LMS and XMSS, the two stateful hash-based schemes, in SP 800-208, published before the FIPS pair.

FN-DSA (Falcon) has been selected and is not yet out as an initial public draft for comment. It is stuck in approvals: submitted, neither approved nor rejected. Meanwhile, fixed-point arithmetic is replacing floating-point. A paper at Crypto this year, and a PQC Forum post from the FN-DSA work proposing edits to the internal initial draft so that it is fixed-point only, are collecting feedback. The expectation stated: Falcon will be standardized; the date is unknown; floating-point may be gone before that happens. A separate limited-signature parameter effort for SLH-DSA was described as an initial public draft already out, feedback being folded in, and a final version close, under a special-publication number that was garbled.

Additional signatures, round 3. Nine further candidates are under consideration, in a third round. The design intent: stateless signatures already include SLH-DSA and ML-DSA; one more general-purpose signature, not necessarily on structured lattices, would add diversity. One drop-in replacement was the original hope. Five or six standards is the world that the tradeoffs produced. There is no planned number of rounds and no quota such as "cut three per round." The on-ramp started with 69 candidates. Many were broken by the public cryptanalysis and dropped immediately. Cutting too wide makes it impossible to bring an algorithm back; cutting too narrow makes the process unbounded. Round 1 resembled earlier competitions: the community broke schemes and the eliminations were easy. Round 2 to round 3 was harder. Diversity of assumptions was preserved on purpose. All four UOV schemes were kept. The largest second-round group was described as MPC-in-the-head and

hash-based-like schemes, and some of those were cut. People are waiting on the standards, so the process is being kept moving as well as fair.

Whether those nine will replace today's defaults: it depends on the use case. ML-DSA was called the general-purpose scheme. SPHINCS+ was not. Another general-purpose standard is still the hope. Some on-ramp candidates have very small keys, and they matter only if the use case is optimizing for that. Otherwise ML-DSA or Falcon is enough. Falcon's performance is better than ML-DSA's, subject to how fixed-point performance actually lands, and Falcon was expected to see more use than a typical on-ramp outcome. FN-DSA and ML-DSA are both lattice-based. An on-ramp scheme on a different assumption was described as a contingency, not as the default.

A chat question asked whether HAWK is inside the nine. The answer given was that the nine do not include HAWK. The same conversation is the candidate-count discussion. The chat correction to record with it: HAWK was withdrawn after an AI attack, so eight remaining candidates may be the more accurate count than nine. The spoken answer and the chat correction disagree. Editorial resolution: NIST selected nine candidates including HAWK in May 2026; HAWK withdrew on 29 July, leaving eight of that selected set. See `https://csrc.nist.gov/Projects/pqc-dig-sig` and `https://hawk-sign.info/`.

NIST was described as not writing use-case-specific standards. At an Ethereum community workshop the previous week, variants of SPHINCS (spoken names along the lines of a "minus" variant and a "G" variant) were being built because stock SPHINCS+ is not tuned for that chain's constraints. Those variants do not inherit a multi-year public cryptanalysis just by being variants, and they may still be widely deployed. The feedback paths named: the NIST PQC standardization conference, where tweaked versions of in-process or already standardized algorithms are presented; the PQC Forum mailing list, where research is announced between meetings; ordinary conference publication; and a pqc-comments mail address. Comment periods attach to draft special publications and FIPS. Round-status reports are reports of decisions already taken and do not have a public comment period. Tweaked schemes often go to another body, including the IETF, rather than into NIST. A new SPHINCS variant might be standardized that way, separately.

SP 800-227 was written because deployers were used to NIST language of key exchange, key agreement, and key transport, while the post-quantum proposals were KEMs, which are a different object. The document explains why they are KEMs, points at already standardized key-agreement schemes, gives hybrid guidance, and describes a NIST-approved way to provide key material. An analogue for digital signatures is in progress. What is missing from the literature is guidance on hybrid, dual, or composite signatures: which combinations are approved. Blanket approval is dangerous when one of the two signatures was not produced by a NIST-approved algorithm, because followers of the guidance would import that risk. IETF and CFRG are working the same problem now. An initial public draft of the signature-side guidance was guessed for 2027, and explicitly not for this year.

HQC was asked about by name. It is in the approvals process. It might not be this year; the hope expressed was that it would be.

On FN-DSA timing: it will not be out in a few days. Whether it is out within 365 days was unknown,

because after the technical draft leaves the writing group it enters layers of approval that have already held it without a yes or a no. No ranking of ML-DSA above FN-DSA, or any required selection order, was known. If such guidance existed, it would already be in a current special publication.

NCCoE. The migration initiative was identified with NCCoE. The standards group works with NCCoE and does not itself run that migration project. NCCoE's early plan was guidance, including sector guidance (financial sector among others): both documents and facilitation among companies. One early white paper's sequence: inventory where cryptographic assets are, then write a plan, with inventory first. The discussion's interest was a taxonomy of real use cases (illustrated as type A, B, and C) and model migrations for each. Contributing a missing category into that NCCoE work was taken as an action. NCCoE material is to be added to the competition repository. Slides from this presentation were to be placed on the session web page.

Composition theory for hybrids. A second presentation argued from error-correction practice: migration is not an algorithm swap, and a hybrid of classical and post-quantum mechanisms is attractive only with a theory of composition. Stacking every known algorithm was rejected, matching the morning's impractical "sign with everything" extreme. Metrics have to score at least three objectives: security that holds up, low operational cost, and security that lasts. Agentic AI attacks change how much trust classical algorithms still deserve. Post-quantum schemes are less vetted. They may cover the quantum attack known today and still fail against algorithms that a large, reliable quantum computer makes practical. Composition can add correlations that a human review misses and that an automated agent finds. The attack surface is not only data in transit or data under computation; it includes the intent of the users, because AI-driven attacks are interactive. Hybrid schemes raise operational cost even when that cost is the price of the security goal. The research loop offered: understand the security and the performance of a composition; optimize for latency, size, or another metric; then re-check that the optimization did not remove the property the composition proof relied on. Evaluation has to be holistic and has to include behavior in real deployments, not only submissions. Exotic purpose-built signatures, isogeny-based cryptography, and code-based cryptography were named as neighboring research, alongside quantum error correction and fault-tolerant quantum computing, which pull in the opposite direction from migration because they are how a CRQC would be built.

The discussion pushed the hybrid past "one classical plus one post-quantum." AI cryptanalysis is moving faster than expected. Highly structured schemes, including ML-KEM, make people nervous even with no public attack. A hybrid worth studying might be elliptic curve plus one post-quantum scheme, or a dual post-quantum hybrid (ML-KEM with HQC was the example) in which the elliptic-curve leg can later be dropped and redundancy remains. The open end of that series — how many schemes get stacked — was acknowledged. It was still treated as worth defining.

Choosing which signature is live, on a chain that can switch. If a signature is used once, even a stateless scheme can encode an OR: at signing time, expose the public key that will be signed against. Someone, the user or a network rule, has to know which algorithm is the secure one. A concrete construction: keep two algorithms that are believed secure. For ECDSA and SPHINCS, hash the ECDSA public key, hash the SPHINCS public key (SPHINCS public keys are already hashed), and

commit to a Merkle root. At signing time, open the branch that matches the current risk judgment. When ECDSA is broken, the pair becomes SPHINCS plus ML-DSA, with ML-DSA hashed. The standing pattern is one expensive scheme that is treated as ultra-conservative, plus one everyday scheme. When the everyday scheme breaks, move. The construction always has at least two. Unresolved: how anyone knows which one is still secure, and how the system fails when users choose badly.

The delay between "we know it is broken" and "the chain stops accepting it" was asked specifically for Bitcoin. A headline break of ML-DSA might be actionable in a second, a day, or only by hard fork. Disbelief, and false belief, are part of the delay. Letting wallets and users decide respects consent and also raises the error rate; it moves the governance problem rather than deleting it. User choice was called attractive and dangerous. Even after ECDSA is broken, some holders will want to keep spending with ECDSA to avoid fees. Bitcoin was described as a setting where that choice can be left with the user, including the harm to users who choose badly. Ethereum was described as more protective of users. TLS was the comparison for a stack that does not leave a bad choice as "your problem.".

The governance contrast: Ethereum decision-making was described as able to make a choice canonical. Bitcoin was described as lacking that path. A hash-based, hash-everywhere roadmap was described as Ethereum's way of retiring agility, on the assumption that hashes remain secure. The Bitcoin construction above does the opposite — it makes agility a user-visible choice — and it still depends on hashes and Merkle trees. If hashes break, that agility mechanism breaks with them. Collision resistance was distinguished from the attacks that would matter: preimages of public keys, or chosen-prefix attacks that yield a valid public key. Those were the attacks the discussion was least worried about. Faith in hash-based schemes was shared across that exchange. It was also flagged as a statement that might age badly.

ML-DSA's place in a Bitcoin transition. One Bitcoin-facing plan had started with ML-DSA as the post-quantum signature. The revised plan: Q-day is a forcing function. Whether it is ten years away or imminent, the merge happens just in time, and a shorter clock raises operational risk. The failure mode to avoid is adopting a scheme and learning it is broken in the same window in which a quantum computer breaks ECDSA. The loss of confidence would make the next migration harder than the first. So SLH-DSA first, as cover for the chance that lattices fail at the same moment as ECDSA, and only then a lattice scheme as the smaller everyday stateless signature, with the hash-based scheme retained as the backup if ML-DSA fails. That is a post-quantum-plus-post-quantum hybrid, and it was called attractive in the room. A write-up of this argument was put in the chat. It is a shared note on crypto agility for cryptocurrency. No URL was spoken. Section 4 of that note was later read back from chat (below).

Two chain families, as evaluation criteria to be ready before the end of the year. The competition's criteria have to cover both families. Differences split into governance and mechanism.

Governance: Ethereum was described as having a settled path, so a chosen change can move quickly. Bitcoin was described as anarchic, slow, and valuable in part because it appears that nobody is in control. That property was treated as something the criteria should not flatten.

Mechanism: Ethereum has consensus-layer signatures (proof of stake) and ordinary transaction

signatures. Bitcoin does not have consensus signatures. Bitcoin is built not to reuse public keys and often does reuse them, sometimes for good reasons.

Three signature jobs, possibly three schemes: consensus signatures, Ethereum transaction signatures, and Bitcoin transaction signatures. One scheme for all three was wanted and was not expected. At least two approaches, possibly three, follow from the tradeoffs.

Solana's constraint is transactions per second. Signatures that are too large pull throughput toward Ethereum's. Once throughput is close, further differences were called more philosophical than technical. Solana's requirement is to hold a TPS level.

On Ethereum, a SPHINCS-family variant on the execution layer can be the same scheme with a hypertree pruned for the consensus layer: same idea, different parameter sets and sizes. If that is rejected, XMSS was what validators were said to want, because anything else is too expensive at their vote rate. Solana validators cast more votes than Ethereum validators, again because of throughput, and want the same rapid signing.

Social consensus was ranked harder than the cryptography. The retreat showed incompatible opinions. AI has widened the argument ("spend more tokens, maybe a better scheme appears"), where previously few people could propose a signature at all. That is both useful and a path into unbounded search. Bitcoin's dispute over what the asset is for will outlast the start of the post-quantum work. Resistance to post-quantum cryptography on Bitcoin was expected to continue. Published timeline targets stated in the room: Ethereum, full migration in 2029; Solana, roughly 2029; Zcash, 2027, with doubt that 2027 is achievable.

Migration is step one. Step two is holders who do not migrate. At an ECC sunset the options named were: burn the coins, freeze them, or leave them for a quantum attacker. All three are technically available. None is socially settled. Lost bitcoin — keys gone, holders dead — was estimated at 10 to 13 percent of the supply. A proposal numbered 361, offering options for that stock, was described as poorly received and still necessary. The migration conversation is incomplete until the post-migration rule for unmoved coins exists.

Auto-research at the retreat, and sig.golf. Retreat time went heavily to signature choice and to an auto-research challenge, sig.golf, aimed at the hash-based signature under consideration. The reported result of spending heavily on model tokens: about a 50% reduction, not only in byte size but also in verification rounds. Two tracks were described together. One is Lean Ethereum: a formalization of the whole stack in Lean. The other is Ethereum-specific optimization of recommended post-quantum schemes, by public challenges that send frontier models at the hash-based design. The current sig.golf scoring was said to be provisional and due to change as lessons land.

The process, as described: every scheme on the leaderboard is generated by a model, with no human authoring of the scheme. A participant receives a prompt, runs an agent, spends tokens, and submits. Formal verification is applied, and it only helps if the property verified is the property that matters. Many entrants are not cryptographers; they have access to a frontier model. The risk is lower than inventing a new hardness assumption, because the entries share a hash-based design, and it is still a

real risk. The drops on the board were attributed to two moves, not to a new assumption: changing the one-time signature at the bottom of the XMSS tree from WOTS to a close variant, and pruning the hypertree. One submission reportedly reduced signature size by about 7%, or almost 10%; the exact figure was not established. After a leap, other agents retarget the new leader and optimize around it. That pattern — a knowledgeable leap, then a swarm — was offered as a possible way to study the hybrid-composition question as well. A family of similar public challenges was said to exist, including one that searches for attacks on hash-based schemes. The site name was garbled.

Implication drawn for the competition: before widespread AI assistance, a NIST standard was what deployers shipped. The present pattern is do-it-yourself signatures per use case. Fragmentation is already visible: Ethereum was described as prepared to take the sig.golf outcome; Bitcoin is looking at a SPHINCS construction that still contains standardized pieces; Solana is using its own XMSS variant on the consensus layer. The competition could be the place that decides "good enough," including a decision that SLH-DSA is good enough, that a smaller signature-budget variant on its way to NIST approval is good enough, and that a higher per-transaction cost is the safer choice. Economics will keep pushing optimization. The scheme itself was called the tip of the iceberg. The hard problems start after the scheme is chosen. Spending the next two years on which scheme to pick was rejected.

A single-hypertree construction aimed at fewer signatures had looked promising until key generation and signing performance were examined. Key generation on one giant hypertree was called too expensive for HSMs, hardware wallets, and phones. A SPHINCS-family construction aimed at the same goals was described as competitive, and as something that might deserve a standardization look, at NIST or elsewhere, if it were not one hypertree. Feedback toward the Ethereum optimization: verification is being weighted too heavily, and the cost lands on key generation and signing. A hardware wallet that cannot store the key would have to rebuild it to sign. Some newer constrained devices may have more memory; that was not established. A scheme both Bitcoin and Ethereum could use would help every other deployer. No historical case of those two networks aligning on a primitive was offered.

Bitcoin's priority order, in reply: verification first, then signature size, with signature size used as a proxy for verification cost. The single-hypertree scheme can have excellent verification and still be unusable if key generation does not run on the target devices. From participation in the auto-research scoring: cutting signature bytes is much harder than cutting verification cycles. One reported outcome was 2 bytes off the signature and more than 100 verification cycles. Whether that is a bias of the models, of the scheme, or of the larger number of verification records was left open, and it may reverse. Grinding (signature blinding used to buy a cheaper verification) was called one of the better levers on the signing side, inherent to the scheme, and safe when it is done correctly.

Workshop deadline. The call for papers had been extended to the Monday of the following week. Short papers, including notes based on this day's discussion, were welcome. The workshop is one day: two or three paper talks, then a closing discussion that is meant to draft the main specification of the competition — which technology is actually in scope. Three or four days remained before the deadline at the time of the session. The conference site associated with that SSR workshop is `https://ssresearch26.umbc.edu/` (PQC workshop call for papers). The captions support the

Baltimore security-research workshop, the extended Monday deadline, and "SSR"; they do not speak this URL. The URL is attached because this slice is the one that discusses that call.

Section 4 of the shared note, read from chat. The question posed: does the scheme let a policy decision change which signing algorithm is accepted, without changing the protocol specification?

Three live Bitcoin positions were distinguished:

Disable insecure schemes. Game theory was said to favor this, because removing vulnerable coins supports the price. It might happen as an explicit change, or as a soft fork that miners run because burning those coins is in their financial interest, even if a public discussion says not to.

Pay-to-Taproot v2: the user opts in, via a soft fork, to having insecure schemes later treated as invalid signatures. Using that template is consent to future disablement.

The Merkle-root construction, identified as BIP 360: the wallet's policy chooses which algorithm to sign with, and the network continues to accept insecure signatures. The note's Section 4, on this reading, is a wallet policy, not a consensus rule that disables algorithms. The author of that reading still expected insecure signatures to be disabled eventually, and BIP 360 itself does not propose the disablement.

A staged migration was asked about: after algorithm A is compromised, accept B only; later introduce C; for a while accept B and C. The mechanism described supports that by spending to a new output. If transaction B spends transaction A, the new output can replace one of the algorithms. That is opt-in and requires the user to generate keys, so adding an algorithm was said to require user consent. Disabling an algorithm does not, on this account, require user consent. The disable-versus-consent choice was flagged for the meeting report and for the evaluation criteria. Which of the three camps wins was left open.

Salvage. An idea previously aired in BGIN: a salvage market for coins whose keys are quantum-exposed, possibly with controlled access to early CRQCs, as an alternative to disablement. Legal reaction in the room, offered expressly as non-advice: salvage of that kind would likely be a serious felony, not a protected salvage right. Moving coins to a burn address for later recovery by a returning owner, or longing the asset and then burning supply with a quantum computer, was distinguished from taking the coins. Taking them was called the legally riskier act. A purpose-built legal category for "ethical" salvage would have to be created. Salvage rights exist in other domains. A lawyer's view was requested. The topic was accepted as future BGIN work, not as a competition rule.

The session closed slightly over time. The evaluation committee was described as now holding both the technical levers and the governance levers to shape a competition aimed first at Bitcoin and Ethereum, and through them at other networks.

Matters not resolved

The candidate count was disputed in the room. The editorial check resolves it as nine selected, including HAWK, then eight after its withdrawal; this is a factual correction, not a new meeting agreement.

Whether FN-DSA's floating-point draft is abandoned in favor of fixed-point only, and whether approval lands inside a year. Unknown. No NIST preference order between ML-DSA and FN-DSA was stated.

Whether HQC clears approvals this year. Unknown.

How a SPHINCS variant built for one chain gets cryptanalysis back into the standards process. Paths exist (PQC Forum, the NIST PQC conference, IETF). No dedicated harmonization venue was identified. Variants do not inherit the parent scheme's review.

Hybrid, dual, and composite signatures: NIST guidance is unwritten; an initial public draft was only guessed for 2027. Combining a non-approved algorithm with an approved one is an open risk. IETF and CFRG are in the same unsettled state.

Classical-plus-post-quantum hybrid versus dual post-quantum hybrid (ML-KEM and HQC as the example), given AI cryptanalysis of structured lattices. Worth studying; no bound on how many legs.

Who decides a signature is no longer acceptable: the user, an opt-in template (Pay-to-Taproot v2), or the network (disable camp, BIP 360 as the user-choice pole). Game theory was said to favor disablement and possibly a miner-led soft fork against a stated community preference. Legal status of a salvage market is open.

How fast Bitcoin can withdraw a broken algorithm (same day, hard fork, or longer). Open, and partly a belief problem.

Whether hashes failing collision resistance would break hash-based signatures or Merkle agility in the way that matters (preimage, chosen prefix). The discussion's confidence was high and explicitly provisional.

One scheme for Bitcoin and Ethereum versus two or three (consensus, Ethereum user signatures, Bitcoin user signatures). Alignment was called helpful and historically unsupported. Solana's TPS constraint may force yet another parameter set, including a private XMSS variant.

Ethereum weighting verification so heavily that key generation and signing miss hardware wallets, versus Bitcoin treating verification as the primary constraint and signature size as its proxy, while still rejecting schemes whose key generation does not run on the target device.

Whether auto-research should keep going. A 50% size-and-verification cut, a 7–10% leap, and a 2-byte versus 100-cycle asymmetry were reported. Formal verification of the wrong property was the stated hazard. The counter-proposal: declare SLH-DSA or a forthcoming smaller-budget NIST variant good enough, pay the fee, and stop spending two years on the scheme.

2027 as a Zcash migration date. Doubted. 2029 for Ethereum and for Solana was reported as those projects' targets, not as a joint commitment.

Burn, freeze, or abandon the 10–13% of bitcoin that cannot migrate, and the coins of users who refuse. Proposal 361 was poorly received. Unsettled.

NCCoE's latest documents were not known in detail. Whether a blockchain use-case category can be contributed is an intention, not a completed input.

High-impact and high-value definitions remain vague even inside the 2030 / 2031 dates.

Disposition of the work

Repository live as a template: `https://github.com/bgin-global/pqc-agility-competition` Aggregation waits on the SSR papers and on the evaluation committee.

Federal transition document: initial public draft 2024, final hoped for around October and stuck in approval. Dates in force for planning: 2030 key establishment, 2031 signatures.

FN-DSA: selected, approval-stuck, fixed-point redesign in public comment on the PQC Forum after a Crypto paper. HQC: in approvals.

Additional signatures: round 3. Editorially corrected count: nine selected including HAWK; eight remain from that set after HAWK's withdrawal.

Signature hybrid / composite guidance: not started as a draft; 2027 was the guess. SP 800-227 is published for KEMs.

NCCoE migration project: inventory-first guidance and sector outreach underway; this community's category input is not yet written.

Chain targets stated, not jointly agreed: Ethereum 2029, Solana about 2029, Zcash 2027.

Ethereum hash-based optimization via sig.golf: about 50% off size and verification rounds, provisional scoring, Lean Ethereum as the formalization track. Leaderboard entries are model-generated.

Bitcoin constructions in play: BIP 360 (wallet chooses the algorithm; insecure signatures still valid) and Pay-to-Taproot v2 (opt in to later disablement), plus a separate disable camp. Proposal 361 on unmigrated coins was rejected in community reception and remains the right problem.

Evaluation criteria for both chain families: due, in the discussion's plan, by the end of this year, as part of standing up the competition.

Actions

Each action is written so that it can be tracked without the rest of the record. Where the room named only a person, the action is kept and the name is omitted.

Add NCCoE material to the competition repository. Follow up the use-case taxonomy and offer a blockchain category if the structure allows.

Post the transition-timeline slides on the session page.

Fold the shared note on crypto agility for cryptocurrency into the archive, and treat Section 4's disable-versus-wallet-policy split as an evaluation criterion.

Submit short notes from this discussion to the SSR / Baltimore PQC workshop. Deadline: the Monday after the session, with three or four days left when this was said. The closing session of that one-day workshop is where the competition's technology scope is to be drafted.

`https://ssresearch26.umbc.edu/` — conference site; PQC workshop call for papers.

Through 2027–2028, the competition timeline from the previous slice still holds (define, then award). Criteria distinguishing governance and the three signature jobs are to be drafted before the end of the year.

Watch FN-DSA fixed-point feedback and the HQC approval. Do not plan on a NIST ranking of ML-DSA over FN-DSA.

Watch IETF and CFRG for composite-signature work. Do not expect a NIST initial public draft on hybrid signatures this year; 2027 was the figure used.

Decide, inside the evaluation criteria, when an auto-research shrink is good enough to freeze, and require that key generation run on the devices that must sign.

Record burn / freeze / abandon, and the salvage idea, as open. Salvage goes to a later BGIN discussion with legal expertise. It is not a competition deliverable.

The committee's near-term focus for multi-chain transition: Bitcoin and Ethereum first.

Coverage

This record covers the published second PQC migration session in Room A, after the tea break, through the lunch announcement. Figures are reported session claims, not independently validated measurements. Technical names are corrected where supported by the recording, chat, or identified primary documentation. Where a name could not be checked against another part of the same recording, it is left unverified rather than guessed. Logistics, introductions, and names of speakers are omitted.

Startup delay ("begin at forty past," one-minute hold), screen sharing, requests to circulate slides, thanks, and the lunch announcement. After the PQC block the room was pointed at AI and blockchain, a break, security-supply-chain governance, a BGIN business meeting, and the evening reception. Those are schedule, not PQC substance. Personal introductions and institutional self-identification are omitted under the Chatham House rule; the technical claims those introductions carried are retained above without the names.

Source clarifications and open checks

NIST IR 8547 and Executive Order 14412 are identified. The earlier instrument rendered "level 10," and the exact applicability of the reported deadlines, still need the transition slides.

NIST SP 800-230 is the identified limited-signature SLH-DSA draft. Its proposed parameter sets must not be described as a final standard.

"Uob" — UOV. "Npc" and "headlike" — MPC-in-the-head and hash-based-like schemes.

SPHINCS variant names "speaks minus" and "speaks G.".

HAWK spelling and the eight-versus-nine count are resolved by the published selection and withdrawal record.

"Book of" — an unnamed WOTS variant that produced one of the leaderboard drops.

The caption may refer to FORS or WOTS. These are distinct hash-based components; the exact construction needs the referenced scheme or slides. Neither name is substituted for the other.

"Hash" as the name of an attack-finding challenge.

Limited-signature "fewer signatures" single-hypertree scheme: no stable algorithm name was spoken.